

Origins and Evolution of Language

Class 5: prerequisites for iterated learning (continued), gene-culture co-evolution

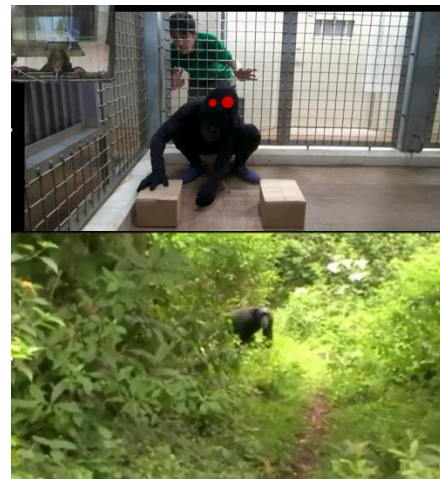
Kenny Smith

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The human package

Somehow, we ended up with

- The ability to infer and reason about the minds of others
- The ability to learn complex grammars
 - capacity for complex vocal learning
 - ability to learn complex sequencing constraints
 - ability to learn compositional meaning-form mappings



Vocal learning

Learning sequencing constraints

Learning compositional meaning-form mappings

Putty-nosed monkey

Pyow = leopard

Hack = eagle

Pyow-hack = move



Arnold, K. & Zuberbuhler, K. (2006). Language evolution: semantic combinations in primate calls. *Nature*, 441, 303



Campbell's monkey

Leopard alarm



Eagle alarm

Boom = not urgent

Zuberbühler K (2002) A syntactic rule in forest monkey communication.
Animal Behaviour, 63, 293–299.

Chimpanzees

Alarm huus: alarm or threat

Waa bark: recruitment call (?)

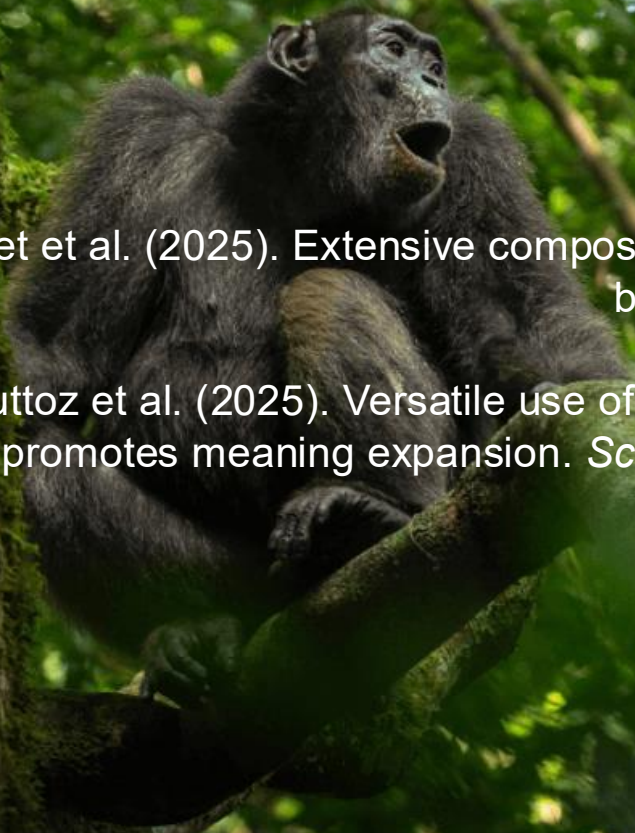
Alarm huu + waa bark = recruitment signal in a dangerous situation (?)

Leroux, M., Schel, A.M., Wilke, C. et al. (2023) Call combinations and compositional processing in wild chimpanzees. *Nature Communications*, 14, 2225.

Extensive and meaningful call combination in chimpanzees and bonobos?

Berthet et al. (2025). Extensive compositionality in the vocal system of bonobos. *Science*, 388, 104-108

Girard-Buttoz et al. (2025). Versatile use of chimpanzee call combinations promotes meaning expansion. *Science Advances*, 11, eadq2879





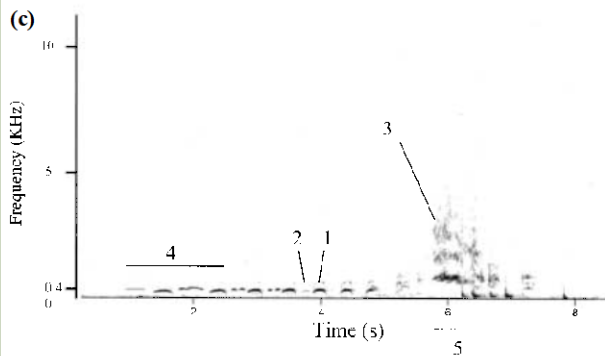
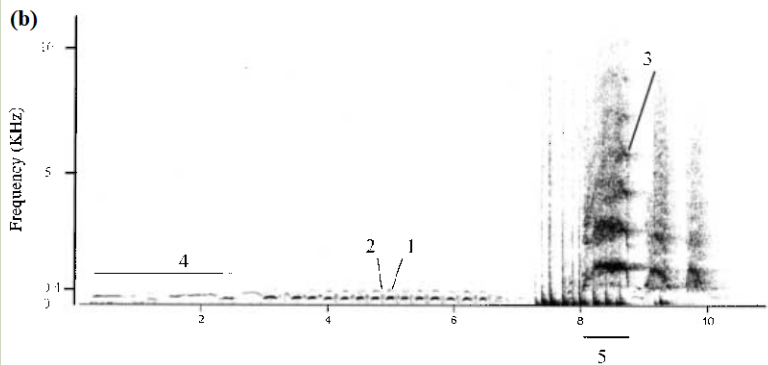
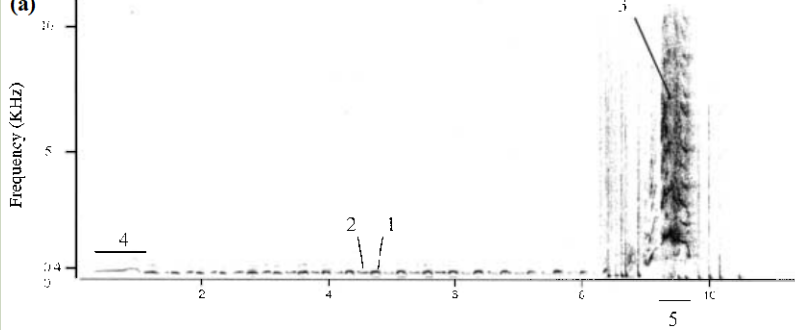
Janie Goodall

Suggestive evidence for learned vocalizations in chimpanzees?



Pant hoots of chimpanzees vary between neighbouring groups

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology*, 110, 221—243.



Crockford et al. (2004): pant hoots of neighbouring groups differ in (e.g.):

- Length of intro (4)
- Peak frequency of screams (3)
- Duration of climax (5)

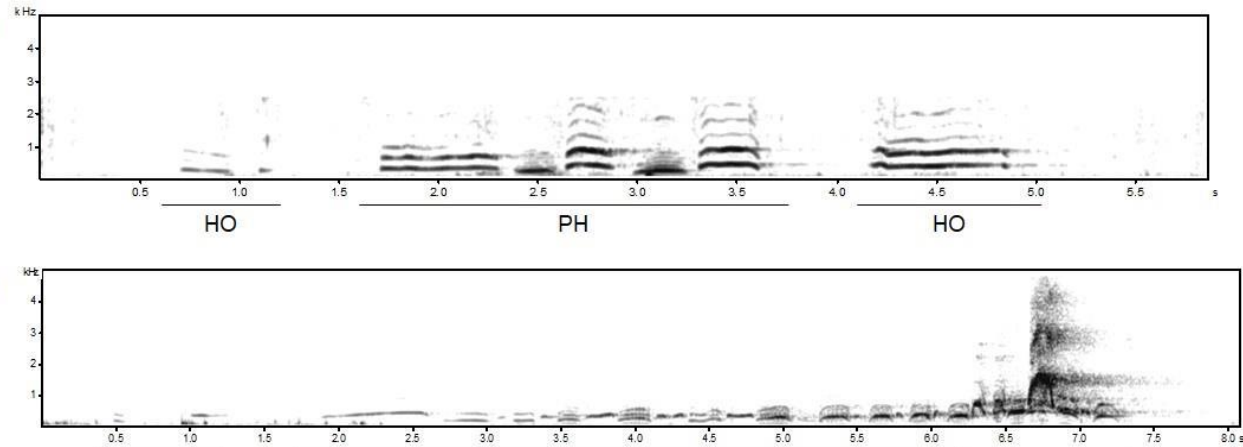
But Desai et al. (2022) fail to replicate in Gombe National Park

- Substantial inter-individual differences, small sample sizes

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology*, 110, 221–243.

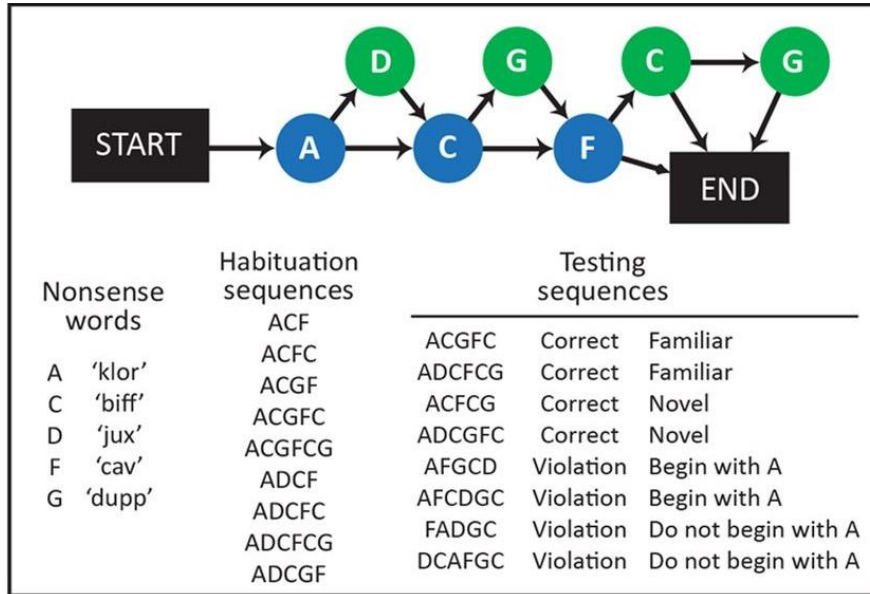
Desai, N. P., Fedurek, P., Slocumbe, K. E., & Wilson, M. L. (2022). Chimpanzee pant-hoots encode individual information more reliably than group differences. *American Journal of Primatology*, 84, e23430.

A lot is not known about call combinations in chimpanzees!

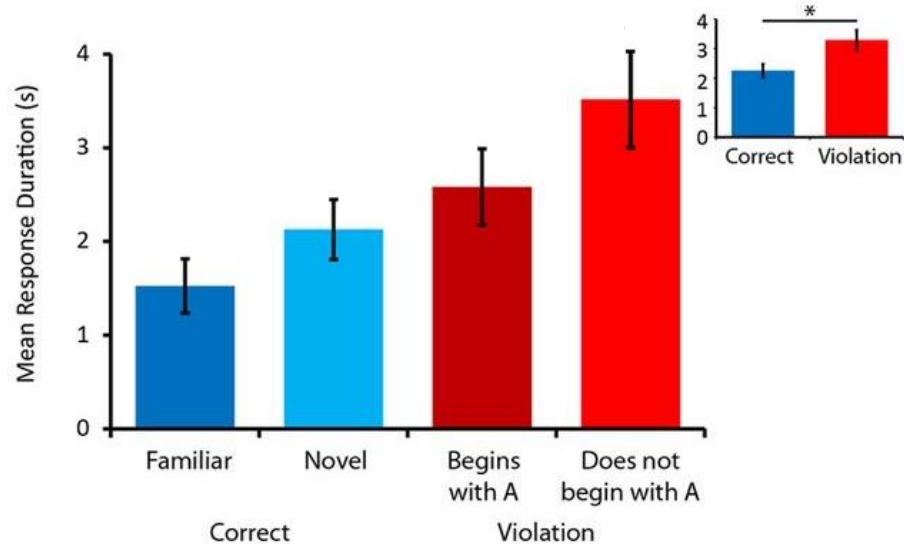
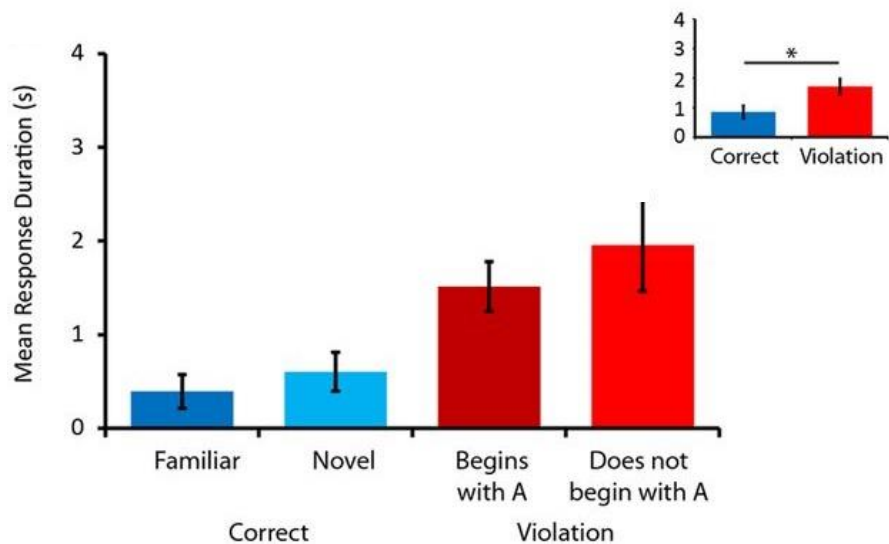


Girard-Buttoz, C., Zaccarella, E., Bortolato, T., Friederici, A. D., Wittig, R. M., & Crockford, C. (2022). Chimpanzees produce diverse vocal sequences with ordered and recombinatorial properties. *Communications Biology*, 5, 410.

Artificial Grammar Learning in non-humans



Wilson, B., Slater, H., Kikuchi, Y., Milne, A., Marslen-Wilson, W., Smith, K., & Petkov, C. (2013). Auditory artificial grammar learning in macaque and marmoset monkeys. *Journal of Neuroscience*, 33, 18825-18835.
 For review see e.g. Petkov, C. I., & Ten Cate, C. (2020). Structured Sequence Learning: Animal Abilities, Cognitive Operations, and Language Evolution. *Topics in Cognitive Science*, 12, 828– 842.



Wilson, B., Slater, H., Kikuchi, Y., Milne, A., Marslen-Wilson, W., Smith, K., & Petkov, C. (2013). Auditory artificial grammar learning in macaque and marmoset monkeys. *Journal of Neuroscience*, 33, 18825-18835.

Non-adjacent dependency learning (pel X rud, vot X jic)

Training

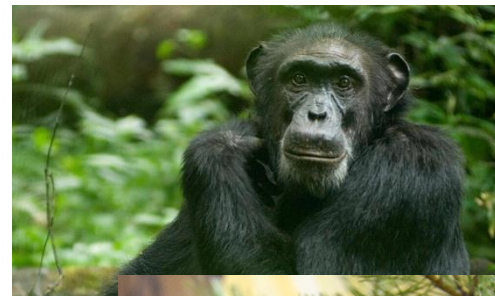
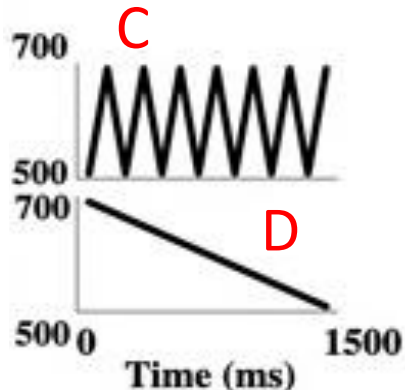
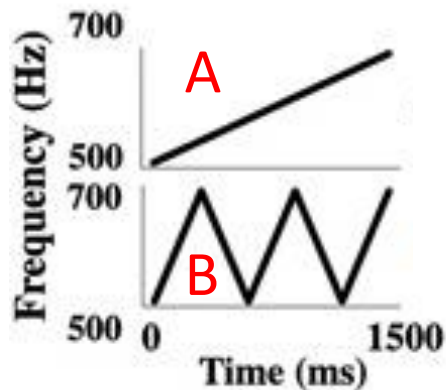
pel wadim rud... vot kicey jic... pel feenam rud...
vot wadim jic... vot puser jic... pel puser rud...
vot kicey jic... vot skiger jic... pel kicey rud ...

Test

pel wadim rud
pel wadim jic?
vot wadim jic
vot wadim rud?

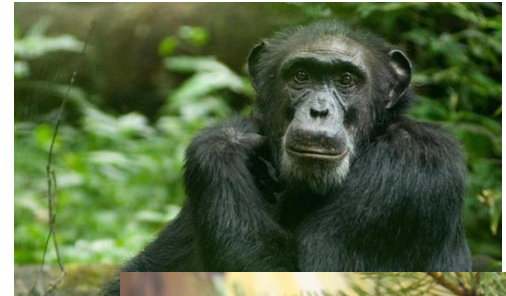


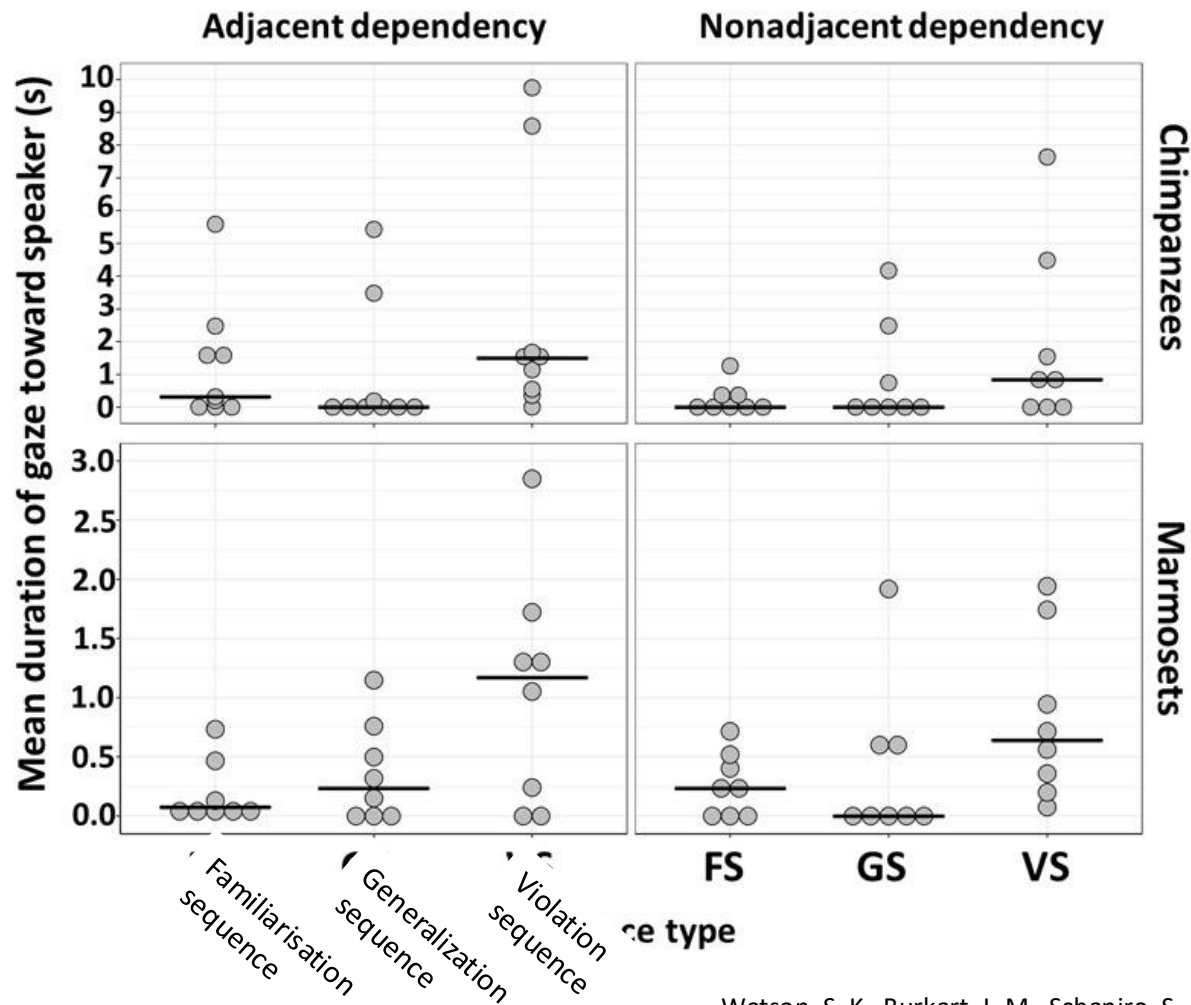
Non-adjacent dependency learning (AXB, CXD)



Non-adjacent dependency learning (AXB, CXD)

Sequence type	Example AD	Example Non-AD
Familiarization	$A_1 - B_7$	$A_1 - X_{24} - B_7$
	$C_5 - D_1$	$C_5 - X_{11} - D_1$
Generalization	$A_{13} - B_{11}$	$A_{13} - X_{115} - B_{11}$
	$C_{10} - D_{16}$	$C_{10} - X_{214} - D_{16}$
Violation	$A_{13} - D_{11}$	$A_{13} - X_{115} - D_{11}$
	$C_{10} - B_{16}$	$C_{10} - X_{214} - B_{16}$





How about learning of **meaningful** sequences?



“ball fetch”
“stick point”

Ramos, D., & Ades, C. (2012). Two-item sentence comprehension by a dog (*Canis familiaris*). *PLoS ONE*, 7, e29689.



“to sugar take decoy”
“to decoy take sugar”

Pilley, J. W. (2013). Border collie comprehends sentences containing a prepositional object, verb, and direct object. *Learning and Motivation*, 44, 229-240.

<https://youtu.be/2Dhc2zePJFE>

Savage-Rumbaugh, E. S., Murphy, J., Sevcik, R., Brakke, K., Williams, S., Rumbaugh, D., & Bates, E. (1993). Language comprehension in ape and child. *Monographs of the Society for Research in Child Development*, 58, 1–252.

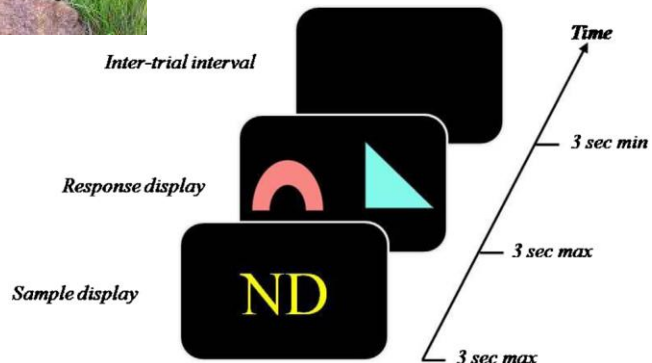
Truswell, R. (2017). Dendrophobia in bonobo comprehension of spoken English. *Mind and Language*, 32, 395-415.

- Could just be ‘semantic soup’ plus smart interpretation?
 - *Cut the onions with your knife*
 - *Put the pine needles in the refrigerator*
- But he can handle reversible events (cf. also Chaser)
 - *Put the tomato in the oil*
 - *Put some oil in the tomato* [Kanzi pours oil in a bowl with the tomato]
- But no strong evidence for **hierarchy**
 - *Give the water and the doggie to Rose.* [Gives dog only]
 - *Give the lighter and the shoe to Rose.* [Gives lighter only]
 - *Give me the milk and the lighter* [Responds correctly]

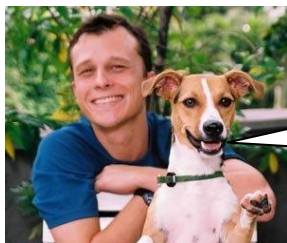


They drive me bananas

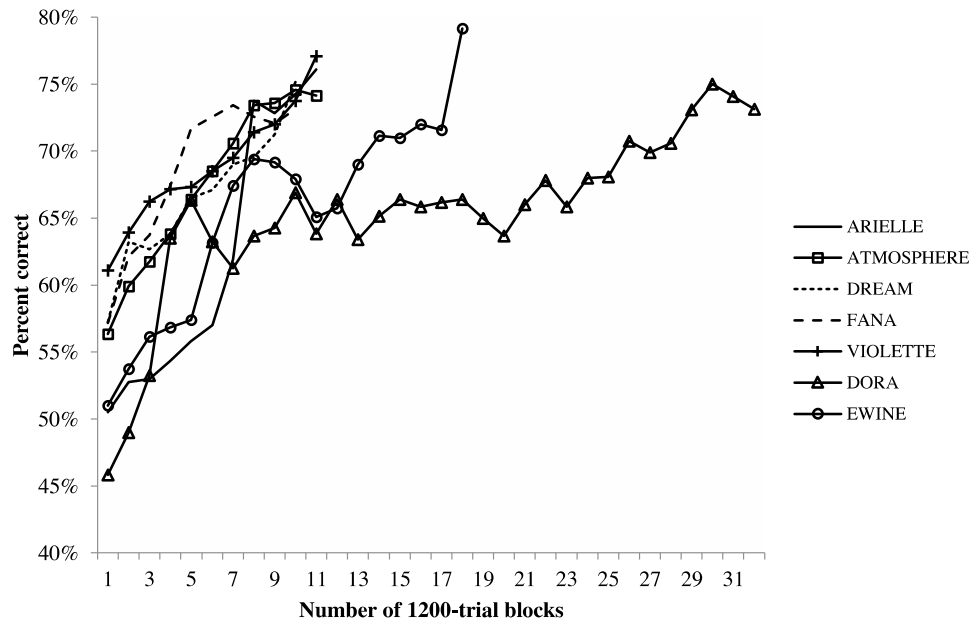
Puzzling failures in (most) baboons



6 letters (3 for shapes, 3 for colours)
3 shapes, 3 colours



Generalization trials are ruff!



Medam, T., & Fagot, J. (2016). Behavioral assessment of combinatorial semantics in baboons (*Papio papio*). *Behavior Processes*, 123, 54-62.

The Last Common Ancestor of chimps and humans

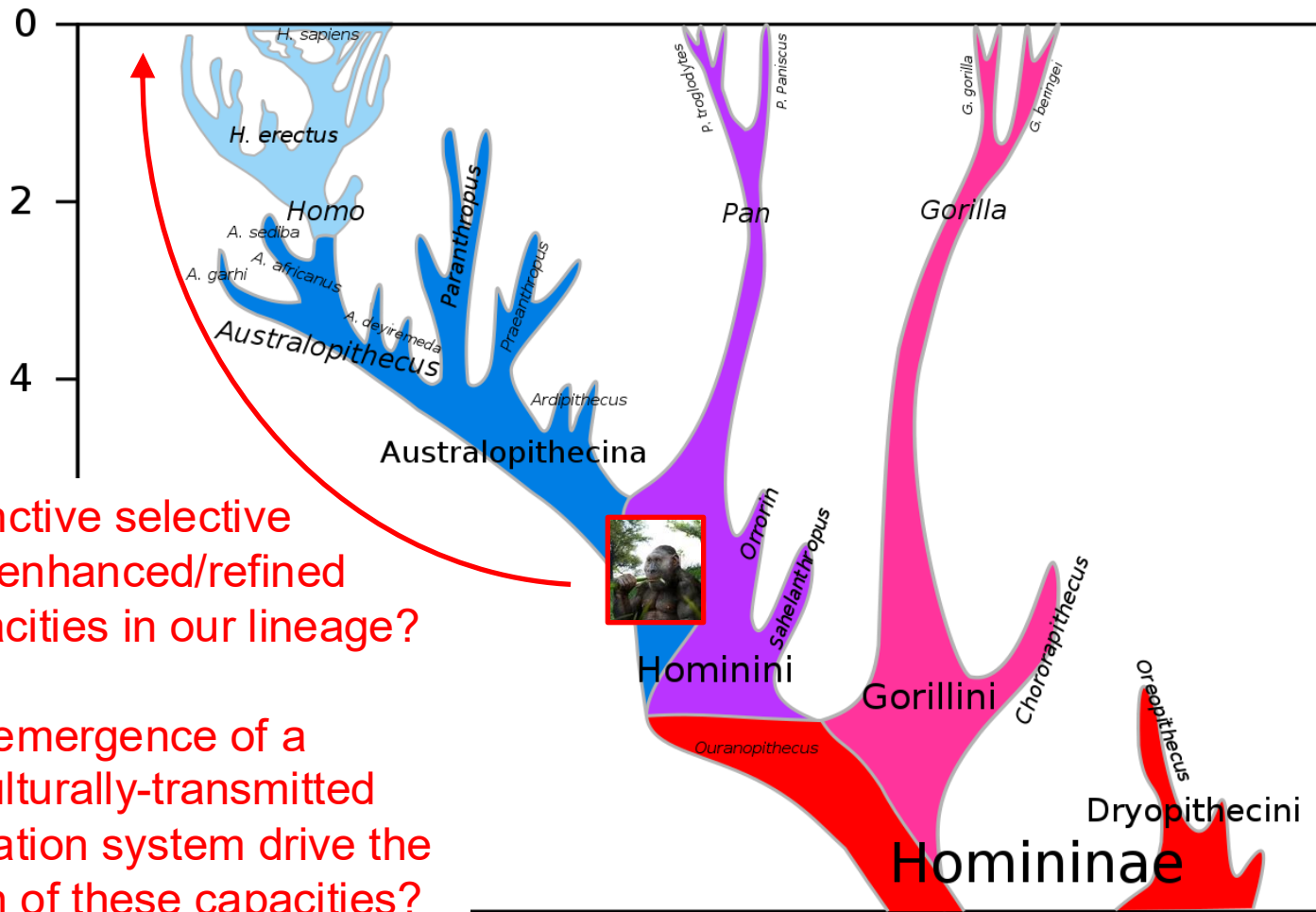


Humans

- The ability to infer and reason about the minds of others
- The ability to learn complex grammars
 - Capacity for complex vocal learning
 - Ability to learn complex sequencing constraints
 - Ability to learn compositional meaning-form mappings

The LCA

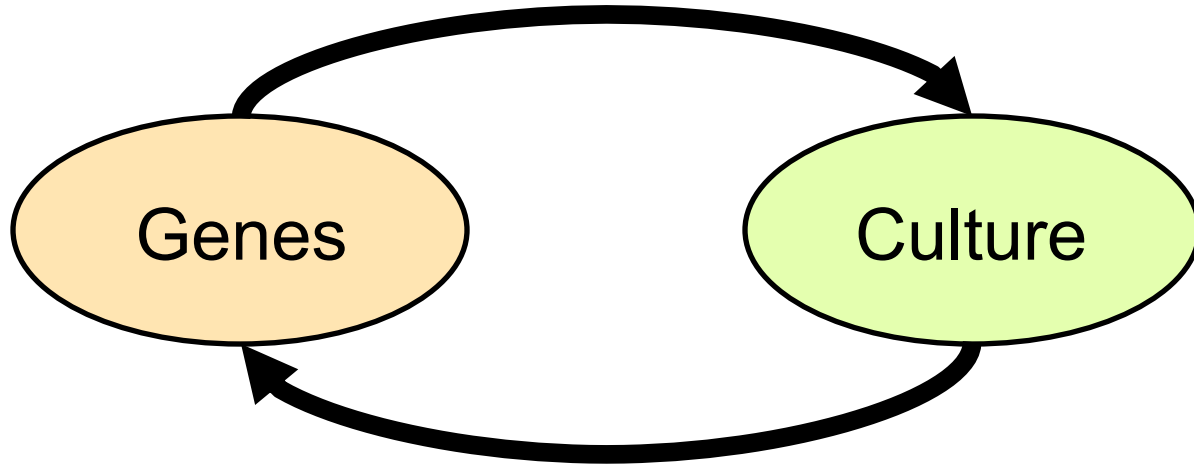
- Some ability to infer and reason about the minds of others
- Some ability to learn complex grammars?
 - Little/no capacity for vocal learning?
 - Some ability to learn sequencing constraints?
 - Some ability to learn compositional meaning-form mappings?



What distinctive selective pressures enhanced/refined these capacities in our lineage?

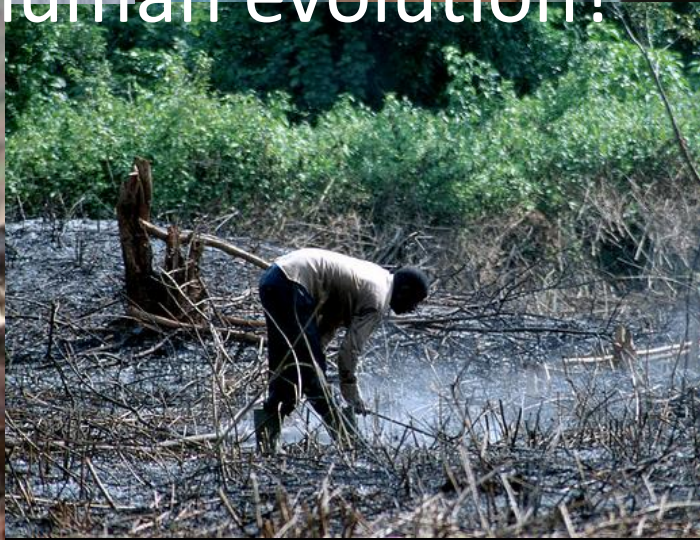
Could the emergence of a “simple” culturally-transmitted communication system drive the elaboration of these capacities?

Gene-culture co-evolution

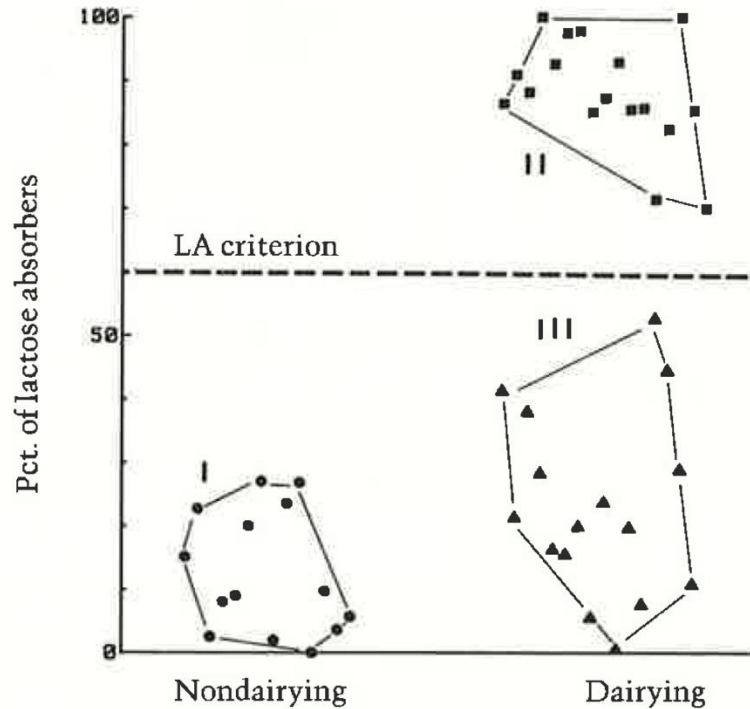




Has culture ended human evolution?

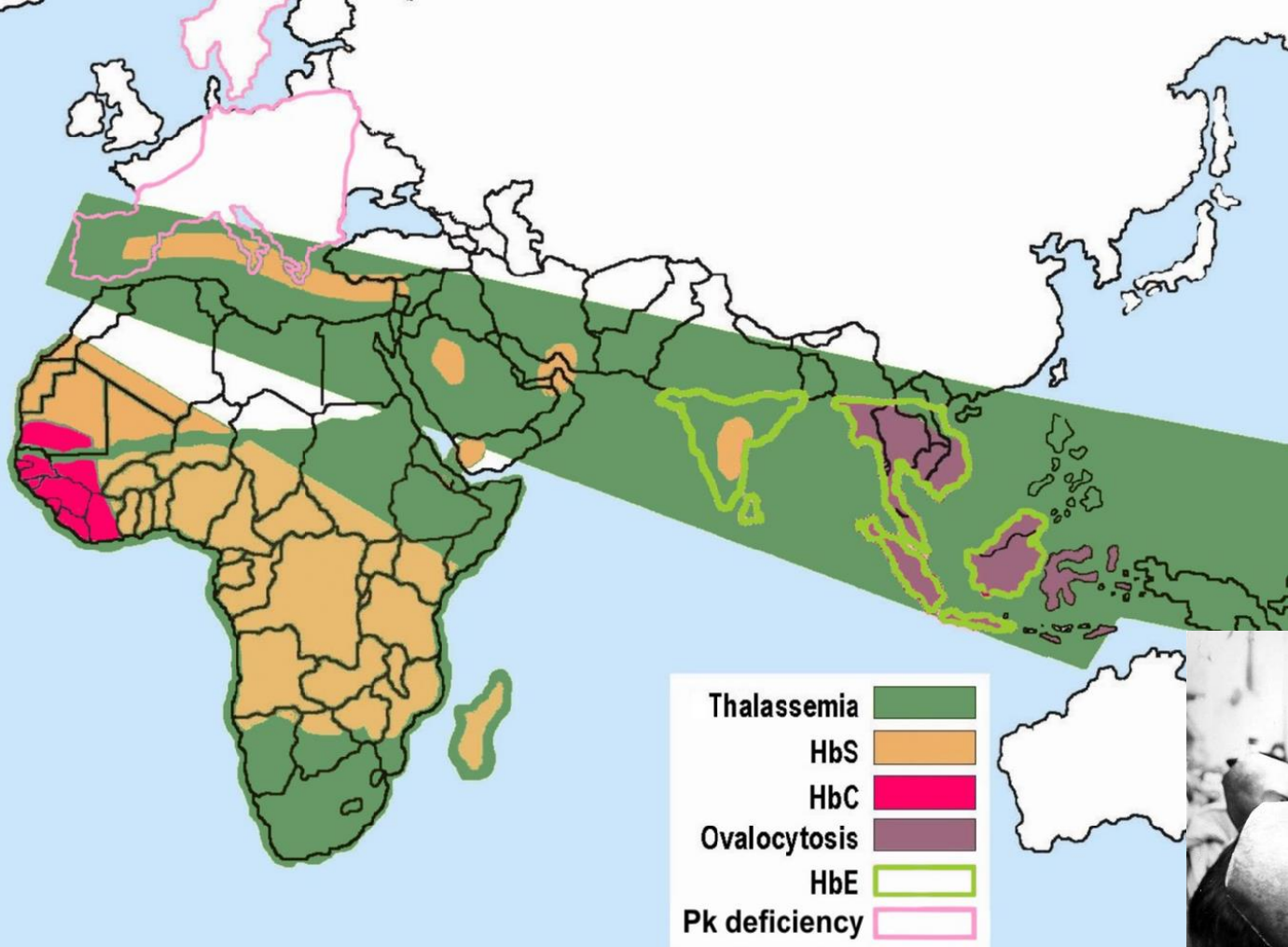


Dairying and lactase persistence

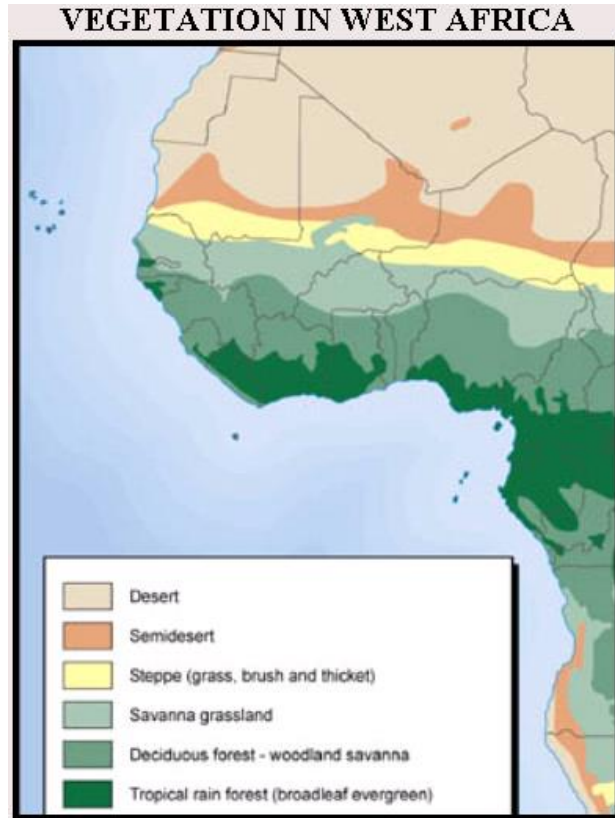


Gerbault, P., et al. (2011). Evolution of Lactase Persistence: an example of human niche construction. *Philosophical Transactions of the Royal Society of London B*, 366, 863-878.

Figure 5.4 from Durham, W. (1991). *Coevolution: Genes, Culture, and Human Diversity*. Stanford, CA: Stanford University Press.



Construction of a malarial niche



Evidence of gene-culture co-evolution

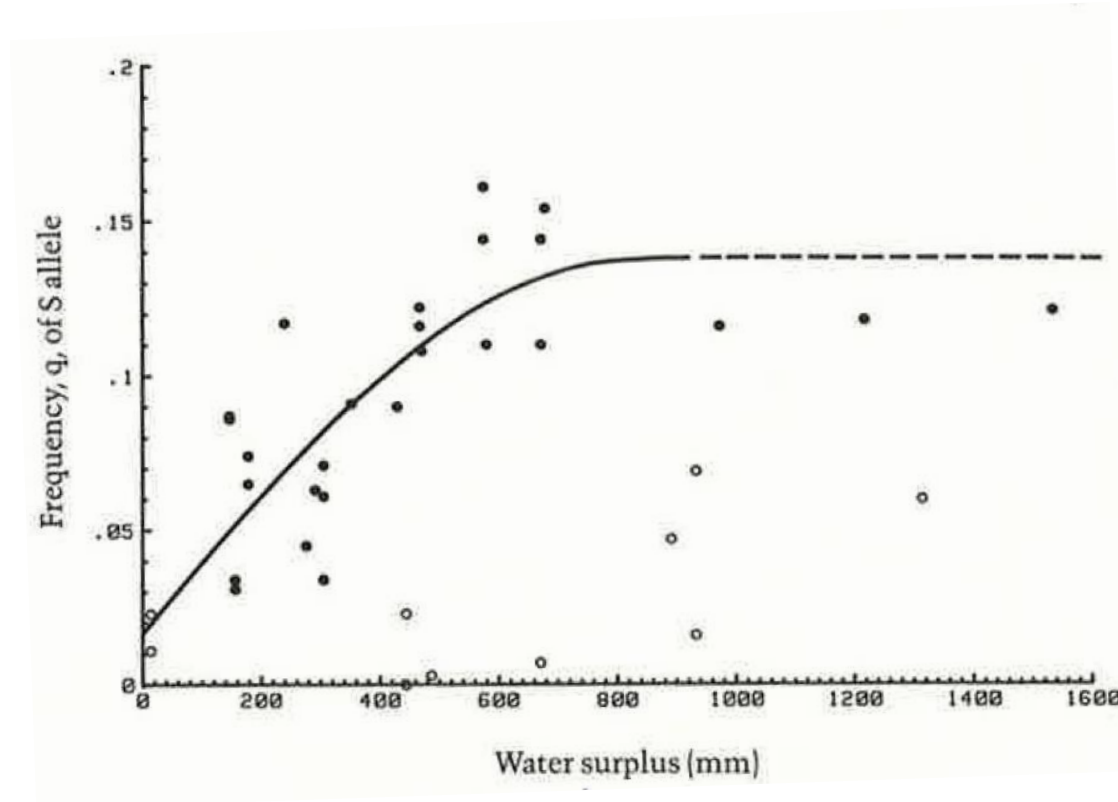
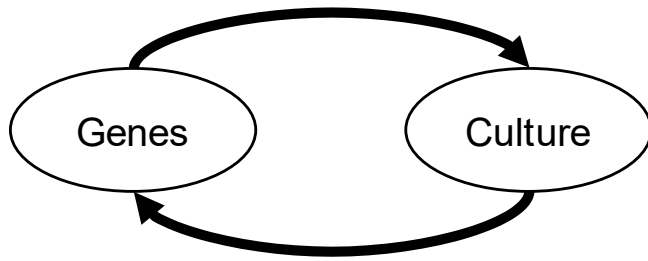


Figure 3.10 from Durham, W. (1991). *Coevolution: Genes, Culture, and Human Diversity*. Stanford, CA: Stanford University Press.

How could this work for language?



How could this work for language?

- Biological adaptations (rudimentary vocal learning, sequence and compositionality learning, mindreading) set scene for cultural transmission
- Cultural evolution begins to create structured (proto-)linguistic system
- Presence of structured, functional language creates/increases selection pressures for language-relevant skills, natural selection ensues
 - Enhanced vocal learning, better sequence/grammar learning, better mindreading
- Enhanced linguistic capacities allow cultural transmission to do more
 - New functions, more complex structures, ...
- And repeat

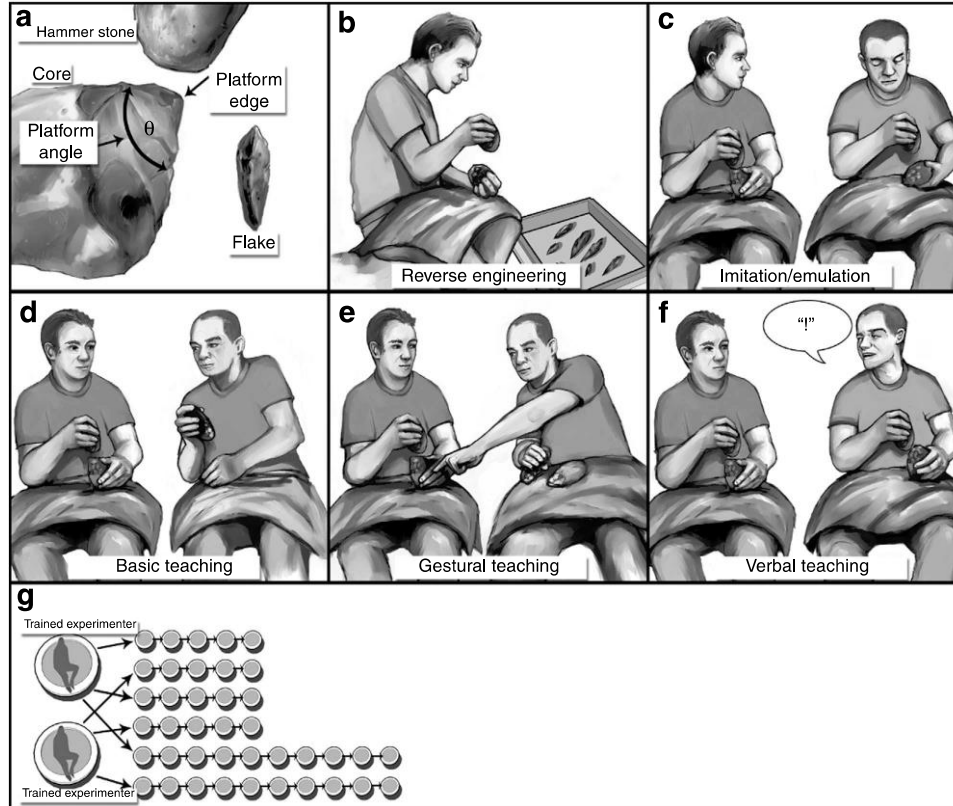
Three examples

- (Experiment) Tool use and language
- (Model) Niche construction in perceptual/articulatory capacity
- (Model) Moving targets and the evolution of innate constraints

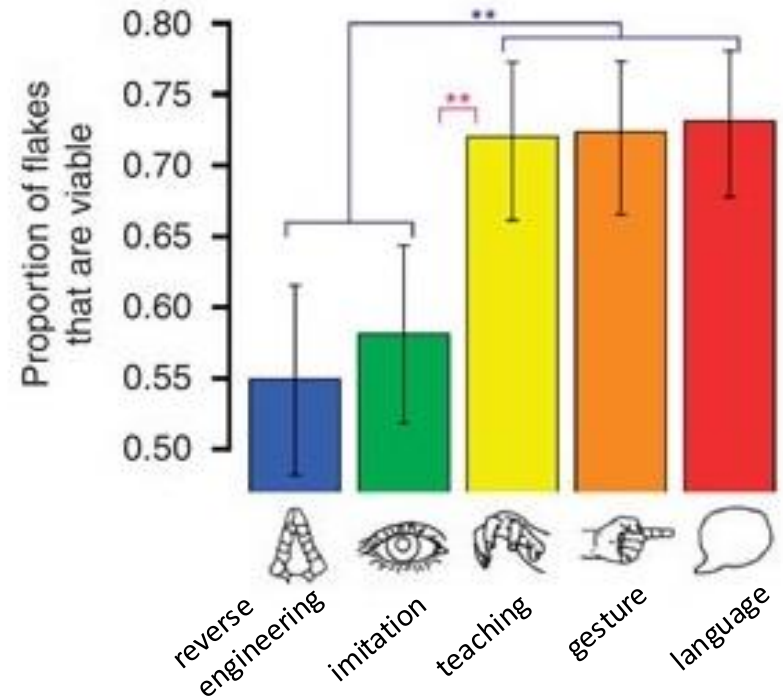
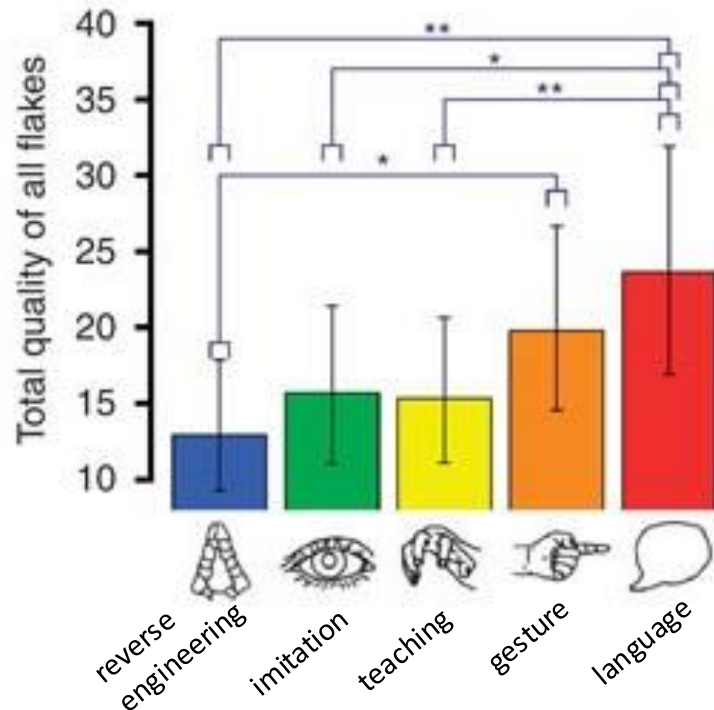
Example 1: could tool use generate selection pressure for language?



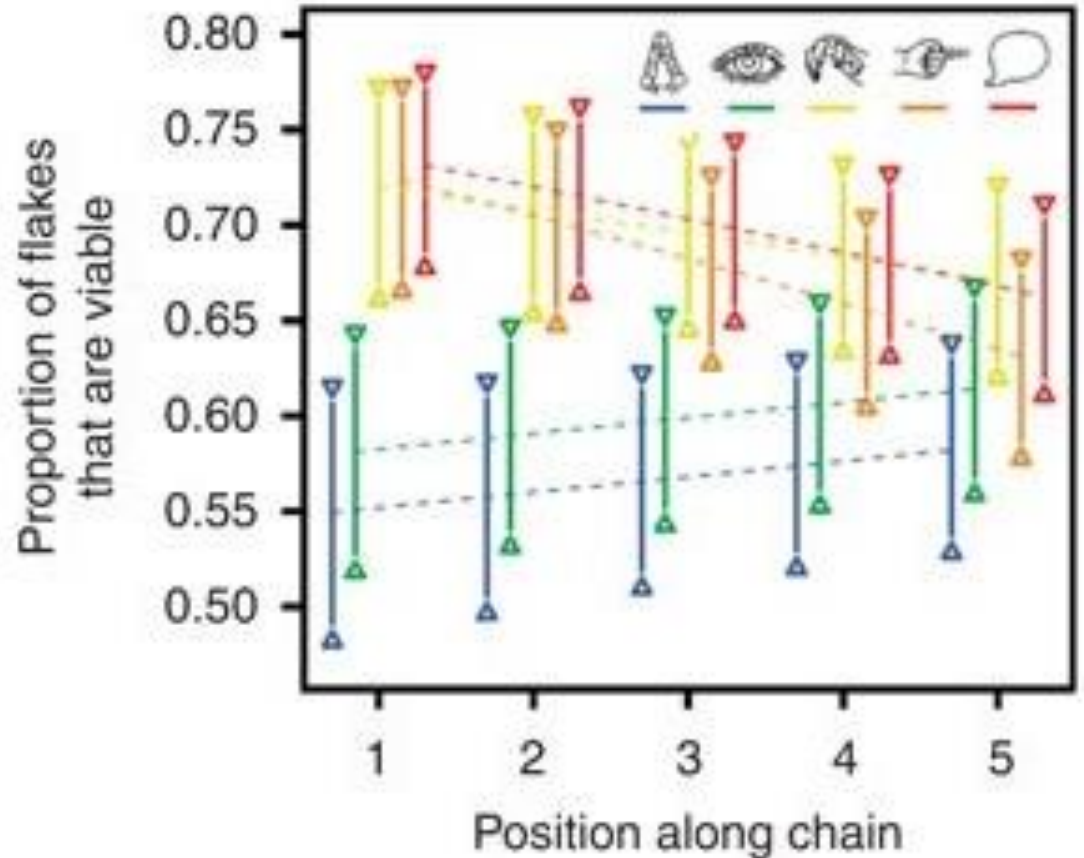
Is language required to preserve stone tool technology?



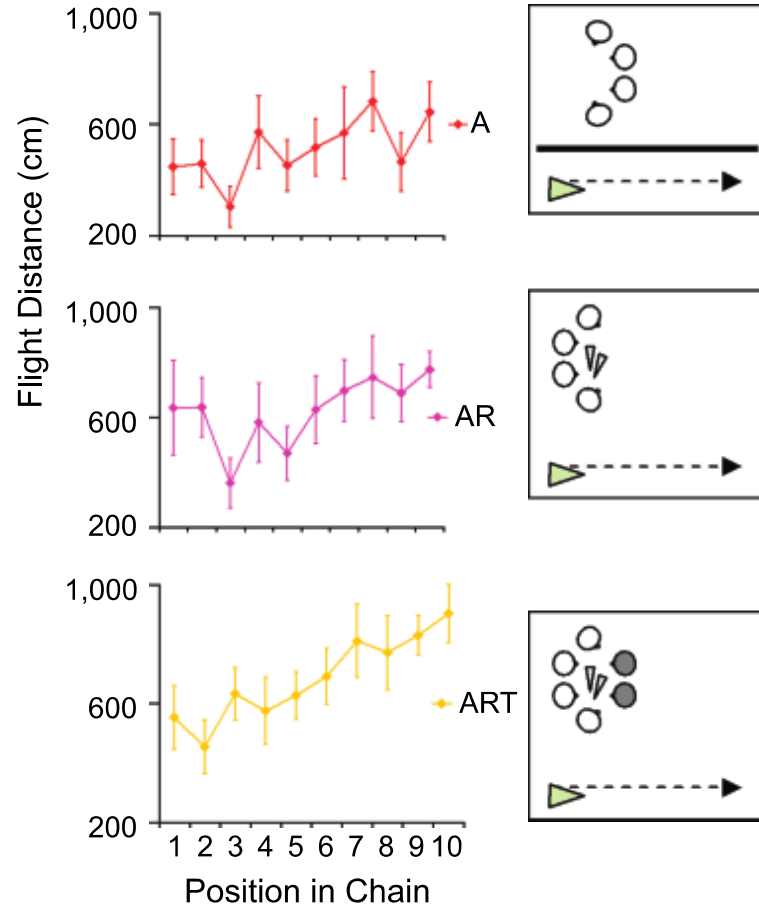
Is language required to preserve stone tool technology?



Is language required to preserve stone tool technology?



Although: no benefit
for teaching in a paper
plane task

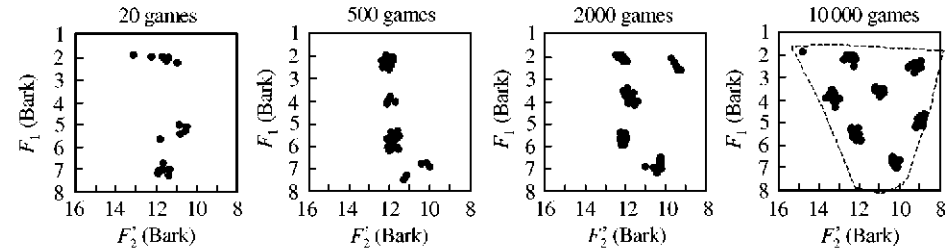
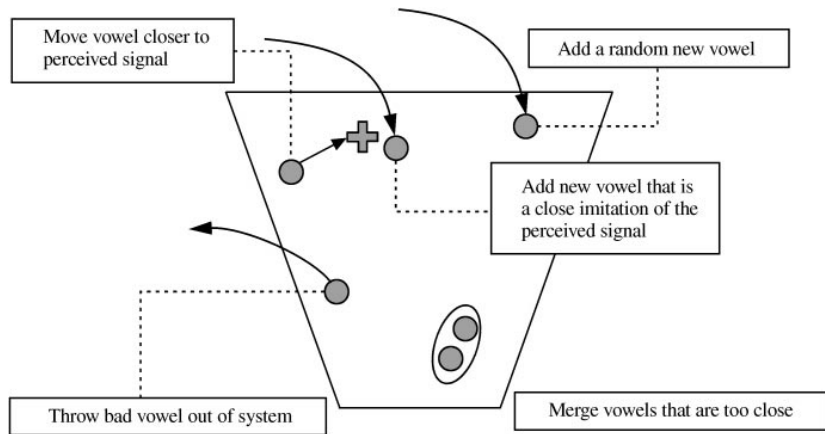


Co-evolution of technology, teaching and language (?)

“our data imply that Oldowan tool making would have created a continuous selective gradient leading from observational learning to much more complex verbal teaching. This process need not have taken place entirely within the Oldowan, but was probably already underway during the Oldowan and likely continued well after, as Oldowan tools continued to be made for hundreds of thousands of years beyond the Oldowan time period.

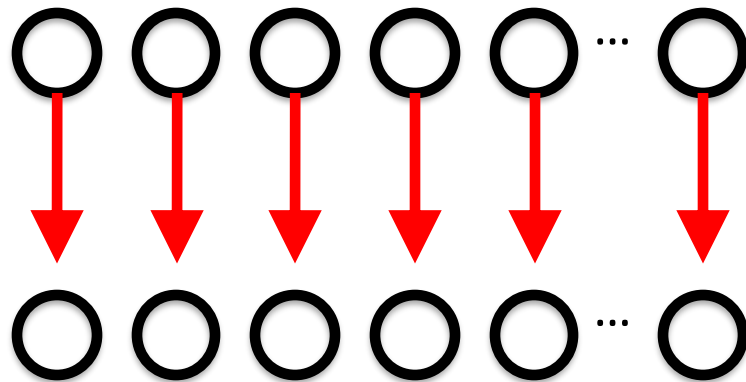
Furthermore, assuming that the transmission of more complex technologies also benefits from more complex means of communication, later technologies would have reinforced the gene-culture co-evolutionary dynamic. Such a process could have lasted for millions of years (and may be ongoing), with more complex communication allowing the stable and rapid transmission of increasingly complex technologies, which in turn generate selection for even more complex communication and cognition, and so forth. Although this places little necessary constraint on when teaching and language may have evolved, our central contribution is to provide evidence that Oldowan tools, produced by hominins since at least 2.5 mya, were involved in this dynamic.” (Morgan et al., 2015)

Example 2: could co-evolution produce increasingly sophisticated linguistic capacities?

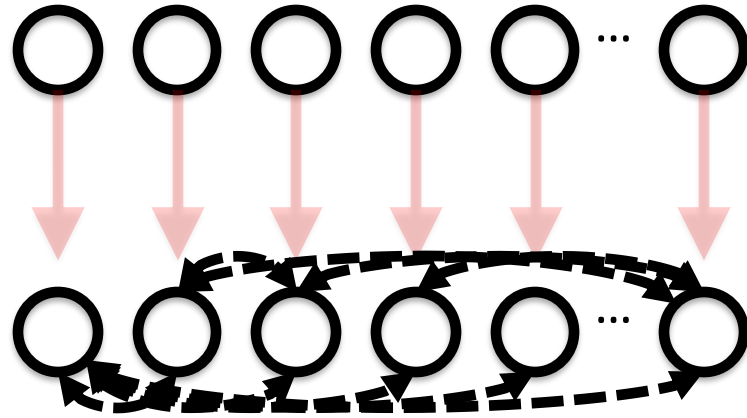


de Boer, B. (2000). Self-organization in vowel systems. *Journal of Phonetics*, 28, 441–465.
de Boer, B. (2016). Modeling co-evolution of speech and biology. *Topics in Cognitive Science*, 8, 459–468.

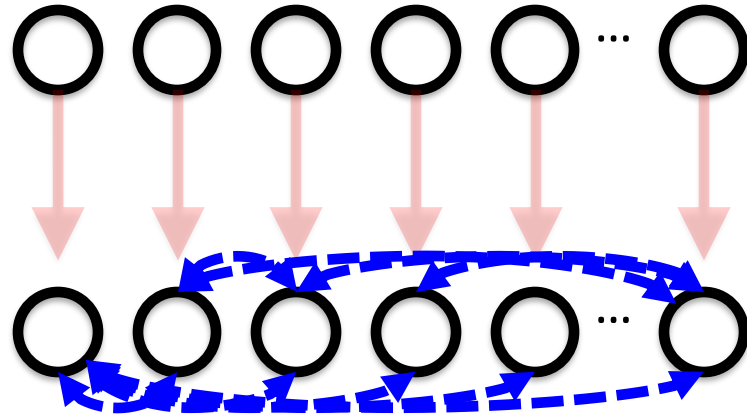
TRANSMISSION



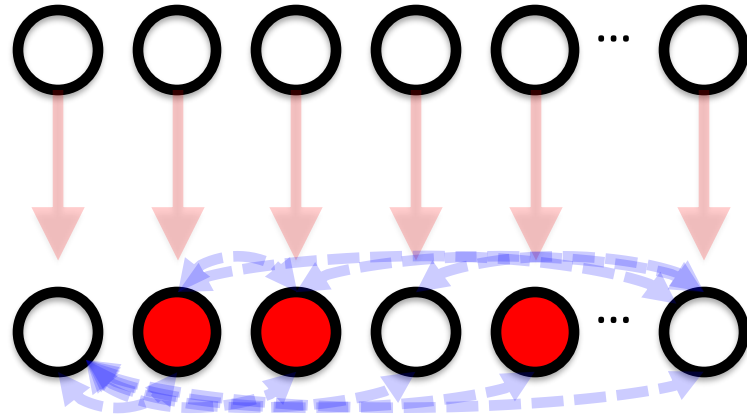
ADAPTATION IN
COMMUNICATION



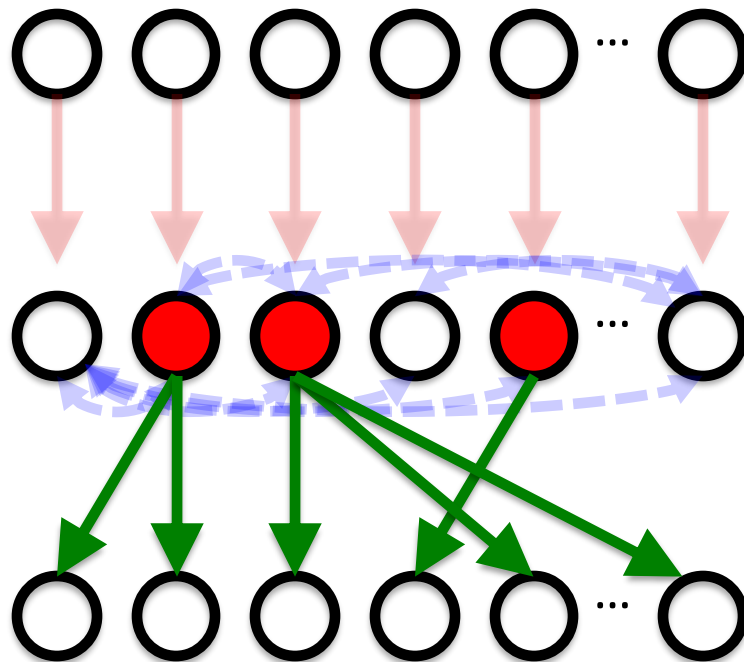
FITNESS
ASSESSMENT



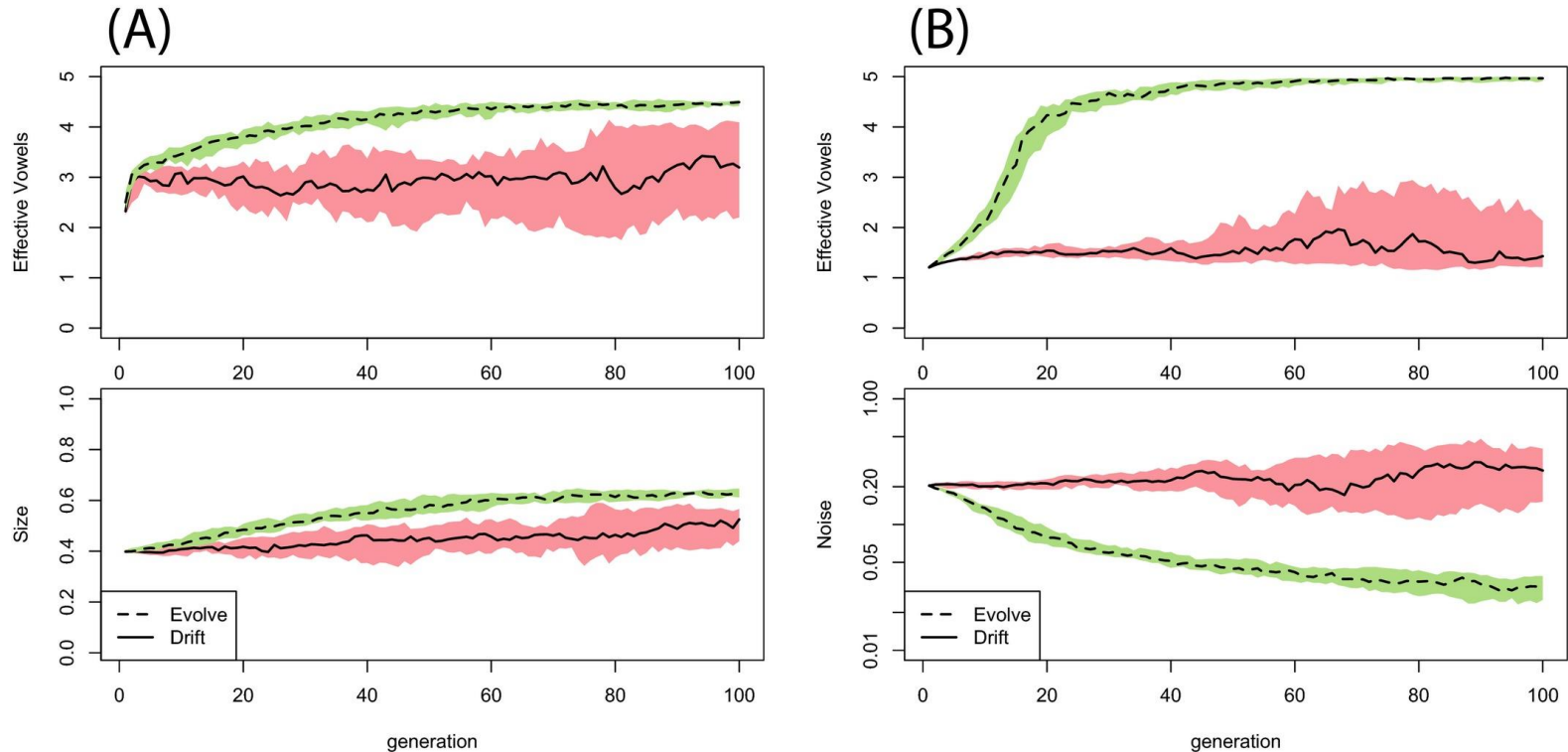
SELECTION



REPRODUCTION
+ MUTATION



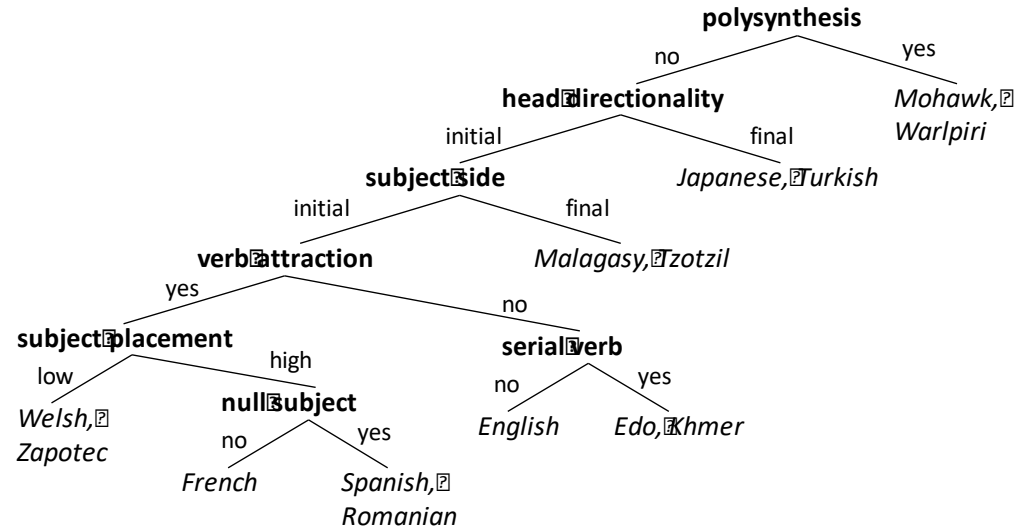
Result: niche construction leads to increasingly complex vowel systems

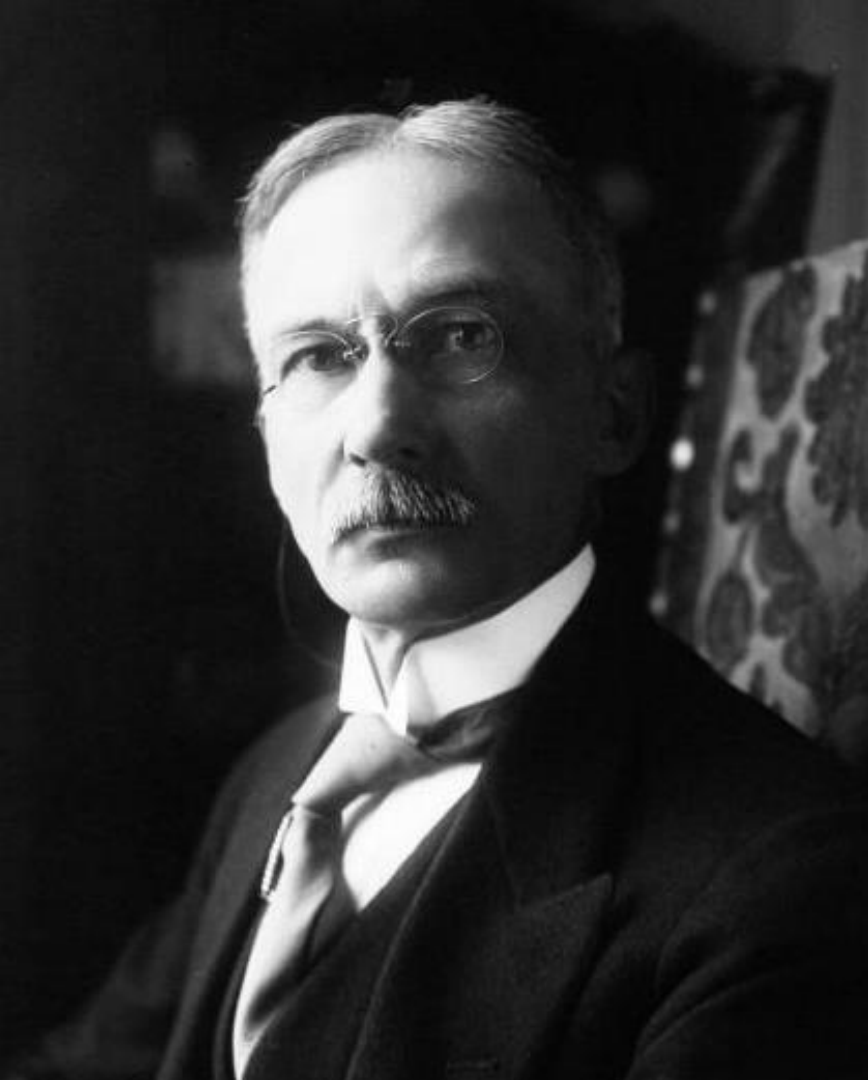


Example 3: could co-evolution produce arbitrary constraints on learning?

Central idea in many linguistic theories: **arbitrary innate constraints** on learning

- e.g. Principles and Parameters





A candidate mechanism: the Baldwin effect (aka genetic assimilation)

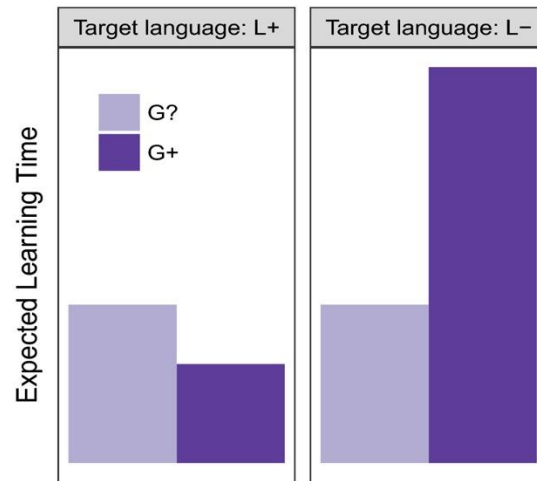
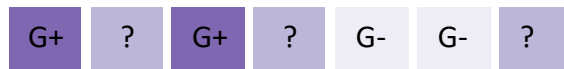
- Behaviour is initially learned
- Learning has some cost (time, **error**)
- Individuals whose genes reduce amount of learning required (e.g. by building in some aspects of the solution) are selected
- Eventually, learning minimized / nativised away

A very simple model of genetically-constrained learning

Language
(series of parameter settings)

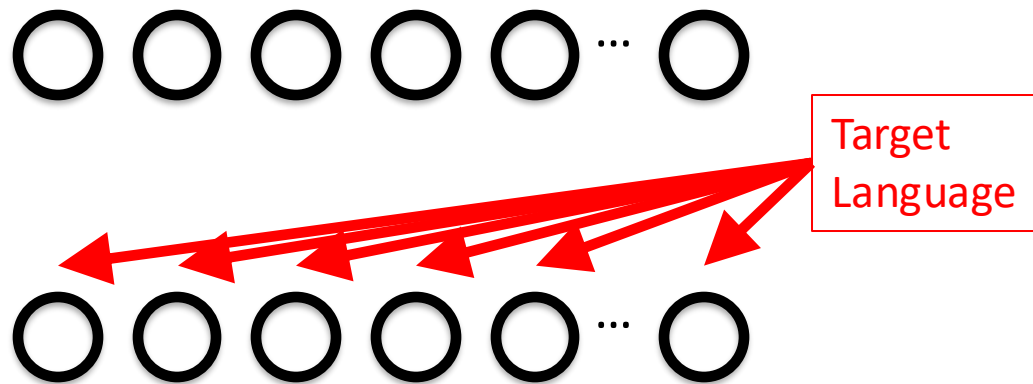


Genes

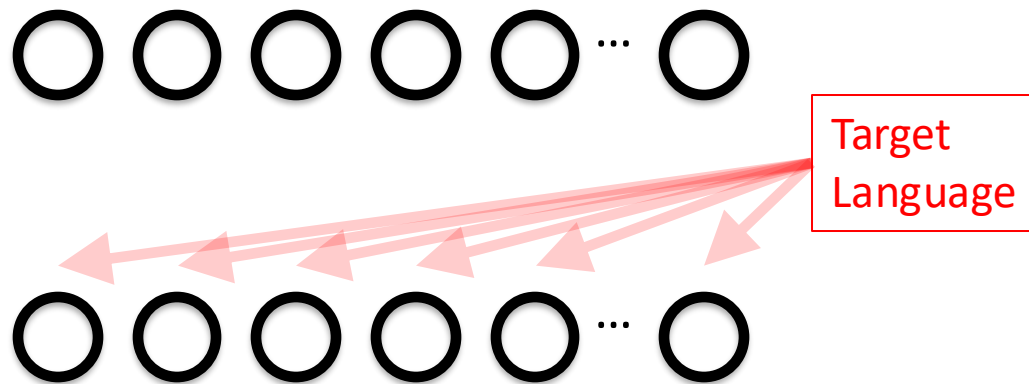


Chater, N., Reali, F., & Christiansen, M. H. (2009). Restrictions on biological adaptation in language evolution. *Proceedings of the National Academy of Sciences, USA*, 106, 1015-1020

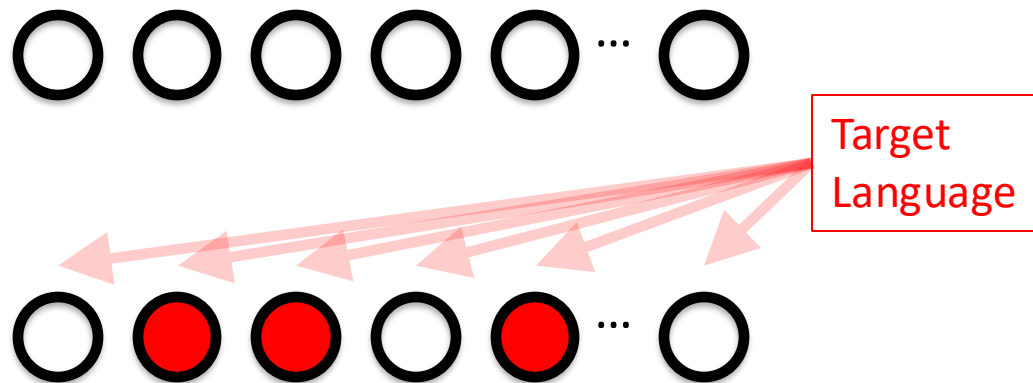
LEARNING



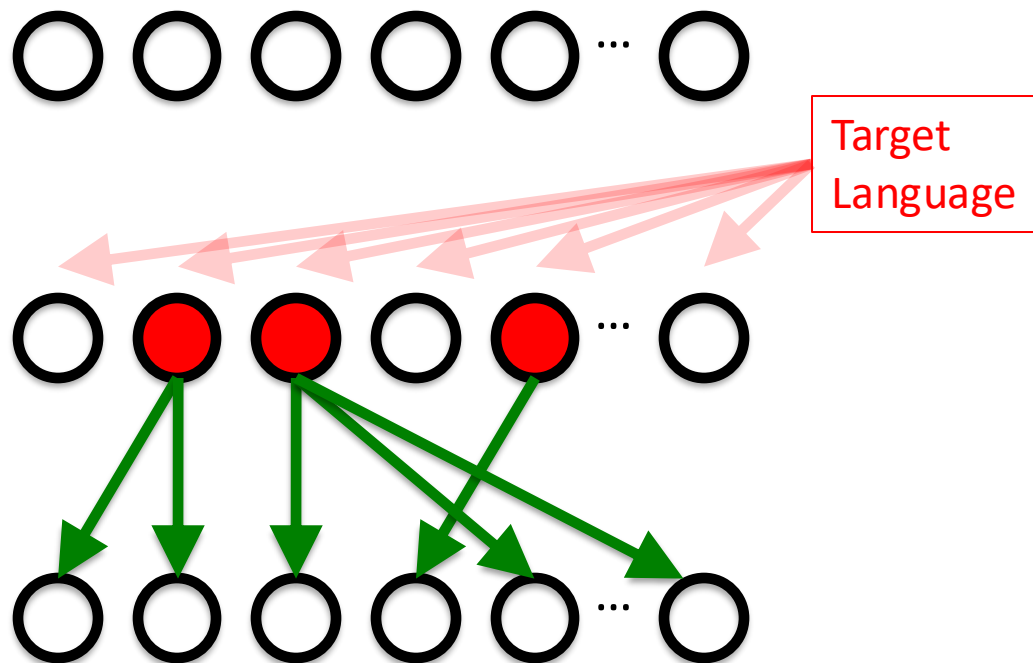
FITNESS
ASSESSMENT
(learning time)



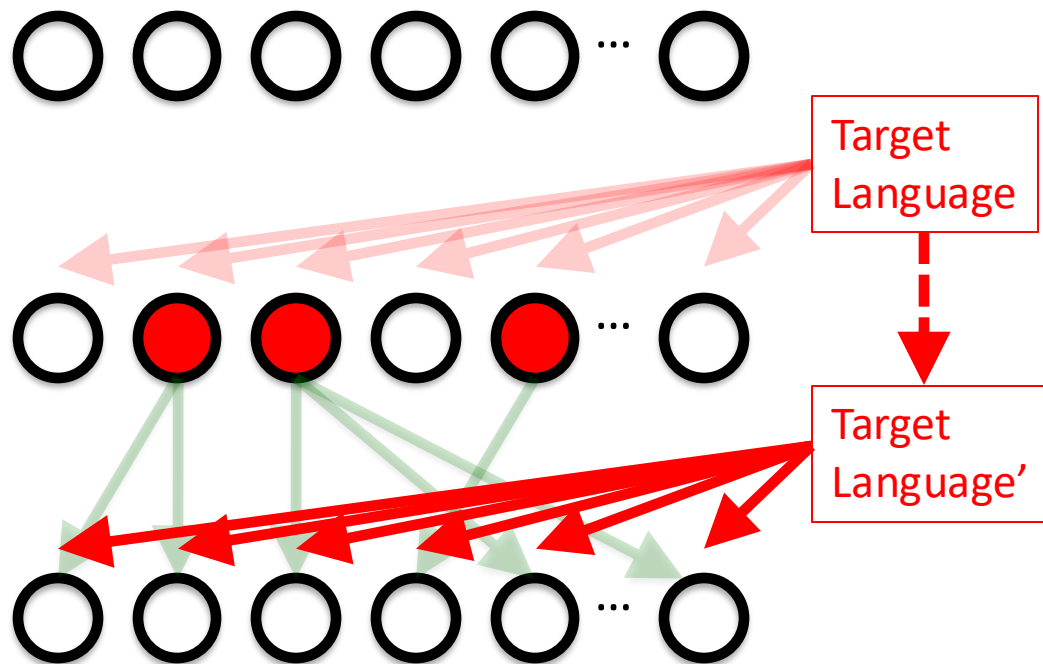
SELECTION

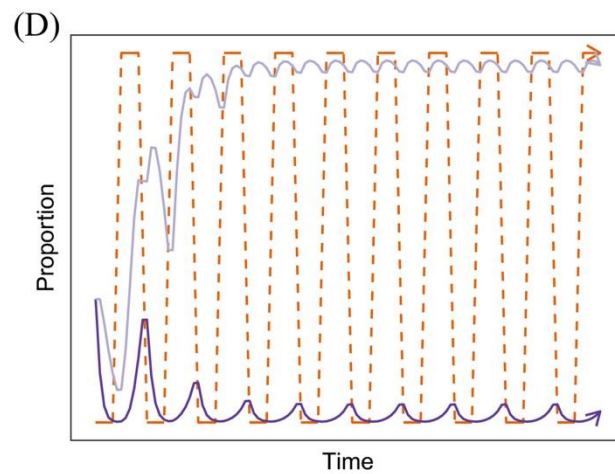
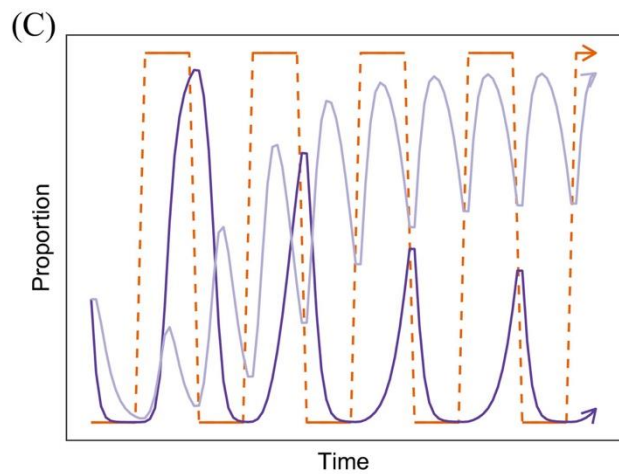
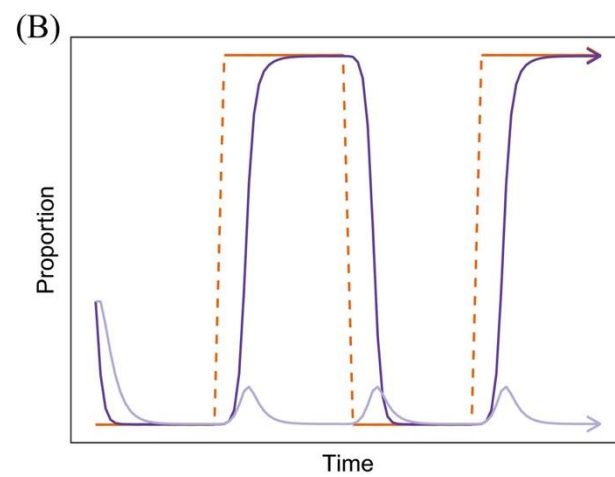
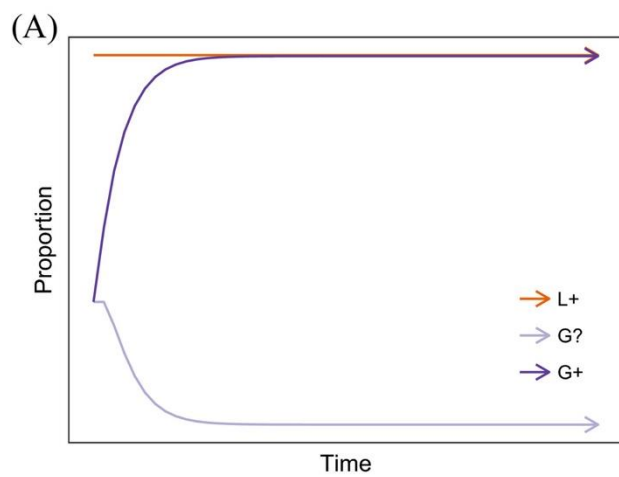


REPRODUCTION
+ MUTATION



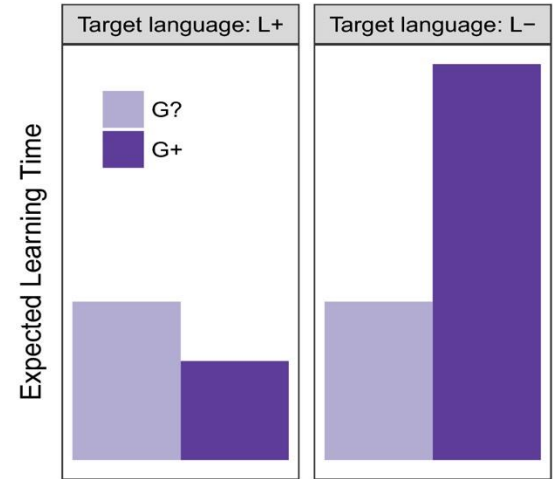
LEARNING





Critique: it's complicated!

- They don't model different strengths of innate bias
- Their learning model disfavors nativisation
- They model language change as a separate process
 - Language changes independent from learner biases
 - No **amplification** of biases by culture



Thompson, B., Kirby, S., & Smith, K. (2016). Culture shapes the evolution of cognition. *Proceedings of the National Academy of Sciences, USA*, 113, 4530-4535

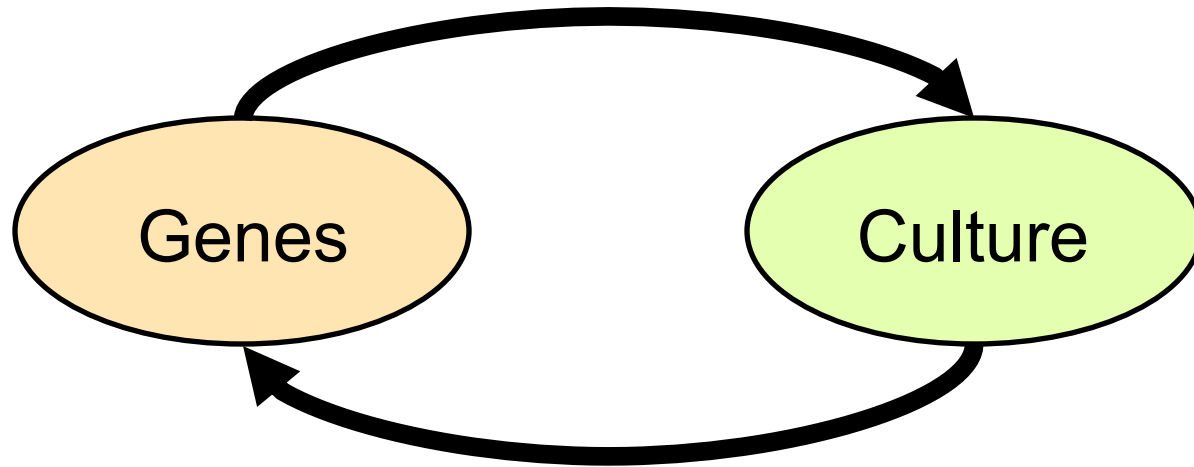
de Boer, B., & Thompson, B. (2018). Biology-culture co-evolution in finite populations. *Scientific Reports*, 8, 1209

Critique: it's complicated!

Two key factors

- What is the link between biological constraints and cultural change?
 - Chater et al. (2009): if the language changes completely independently, biology has a hard time tracking culture
 - If culture amplifies genetic biases, biology can have a **rapid** effect on culture
- How transparent is the link between genes and behaviour to selection?
 - Can selection identify “good” genes?
 - Chater et al. (2009): yes, through learning cost for “wrong” biases
 - Thompson et al. (2016): often not, because learning **masks** genetic biases

How could this work for language?



Course wrap-up

- Language is transmitted via repeated **learning** and **use**
- Fundamental design features of language emerge naturally from these processes
- Some of the necessary cognitive capacities may exist, in some form, in our closest living relatives
- The presence of a culturally-transmitted communication system may have generated selection pressures further refining those capacities

